

Assignment Code: DA-AG-007

Statistics Advanced - 2| Assignment

Instructions: Carefully read each question. Use Google Docs, Microsoft Word, or a similar tool to create a document where you type out each question along with its answer. Save the document as a PDF, and then upload it to the LMS. Please do not zip or archive the files before uploading them. Each question carries 20 marks.

Total Marks: 100

Question 1: What is hypothesis testing in statistics?

Answer:

Question 2: What is the null hypothesis, and how does it differ from the alternative hypothesis?

Answer:

Question 3: Explain the significance level in hypothesis testing and its role in deciding the outcome of a test.

Answer:

Question 4: What are Type I and Type II errors? Give examples of each.

Answer:

Question 5: What is the difference between a Z-test and a T-test? Explain when to use each.

Answer:

Question 6: Write a Python program to generate a binomial distribution with $n=10$ and $p=0.5$, then plot its histogram.

(Include your Python code and output in the code box below.)

Hint: Generate random number using random function.

Answer:

Question 7: Implement hypothesis testing using Z-statistics for a sample dataset in Python. Show the Python code and interpret the results.

```
sample_data = [49.1, 50.2, 51.0, 48.7, 50.5, 49.8, 50.3, 50.7, 50.2, 49.6,  
               50.1, 49.9, 50.8, 50.4, 48.9, 50.6, 50.0, 49.7, 50.2, 49.5,  
               50.1, 50.3, 50.4, 50.5, 50.0, 50.7, 49.3, 49.8, 50.2, 50.9,  
               50.3, 50.4, 50.0, 49.7, 50.5, 49.9]
```

(Include your Python code and output in the code box below.)

Answer:

Question 8: Write a Python script to simulate data from a normal distribution and calculate the 95% confidence interval for its mean. Plot the data using Matplotlib.

(Include your Python code and output in the code box below.)

Answer:

Question 9: Write a Python function to calculate the Z-scores from a dataset and visualize the standardized data using a histogram. Explain what the Z-scores represent in terms of standard deviations from the mean.

(Include your Python code and output in the code box below.)

Answer: